Task 05: Networking and Security Operations with SIEM, Forensics, and Traffic Analysis

Step-by-Step Process & Findings Report

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1. Subnet Design & Network Topology

1.1 Objective

Design an efficient subnet for a small office network supporting 20 internal devices while ensuring scalability, proper IP utilization, and ease of management.

1.2 Step-by-Step: Subnet Calculation

1. **Determine the number of required hosts.** The office needs addresses for 20 devices (PCs and printers). We must also reserve 1 address for the default gateway router, resulting in a minimum of 21 usable addresses.
2. **Choose the smallest CIDR block.** The next power of 2 above 21 is 32 (2^5 = 32). This gives us a /27 subnet (32 − 5 = 27 network bits), which provides 30 usable host addresses — enough for all devices plus room to grow.
3. **Calculate the subnet mask.** A /27 prefix means the first 27 bits are the network portion. Converting to decimal: 11111111.11111111.11111111.11100000 = 255.255.255.224.
4. **Determine the address ranges.** Network Address: 192.168.1.0 | First Usable: 192.168.1.1 (gateway) | Last Usable: 192.168.1.30 | Broadcast: 192.168.1.31.
5. **Assign the gateway address.** Reserve 192.168.1.1 for the router (default gateway). Devices are assigned 192.168.1.2 through 192.168.1.21 (20 devices), leaving 192.168.1.22–.30 free for future expansion.

1.3 Subnet Summary Table

Parameter	Value
Network Address	192.168.1.0/27
Subnet Mask	255.255.255.224
Total IP Addresses	32
Usable IP Addresses	30
Default Gateway	192.168.1.1
Usable IP Range	192.168.1.2 – 192.168.1.30
Broadcast Address	192.168.1.31
CIDR Justification	/27 chosen for 20 devices + scalability

1.4 Network Topology Diagram

The network uses a star topology: a router (gateway) connects to a Layer 2 24-port switch, which runs out to all 20 office devices.

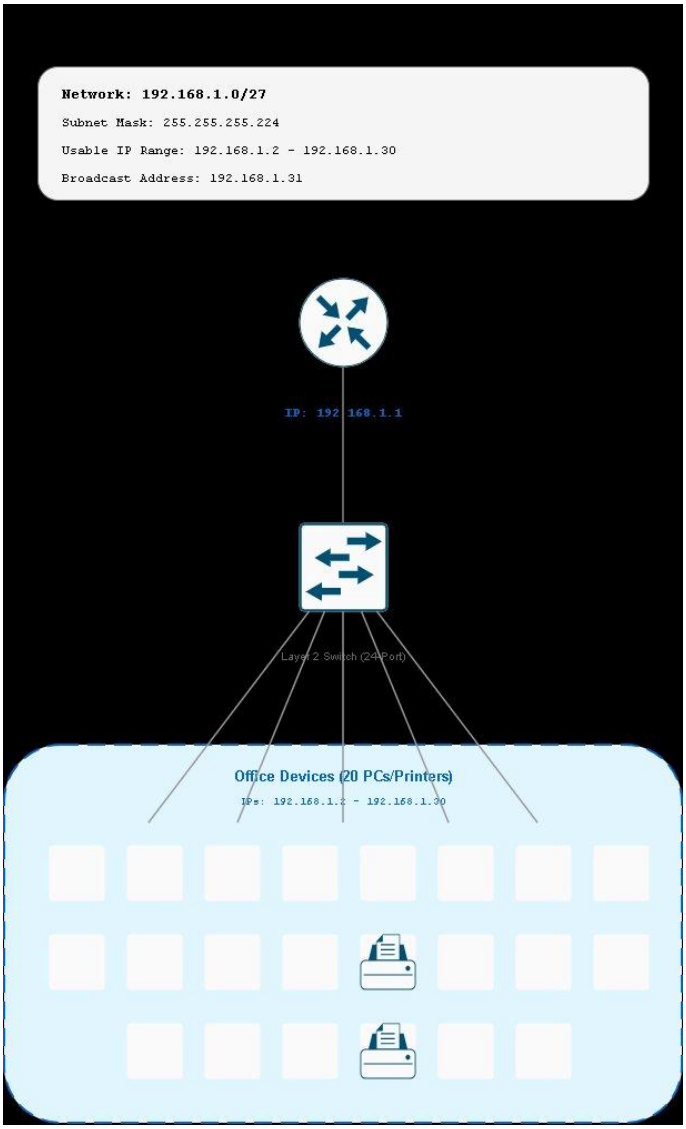


Figure 1: /27 Subnet – Star Topology with Router, L2 Switch, and 20 Devices

1.5 Design Justification

- A /27 subnet avoids wasting addresses while comfortably fitting 20 devices plus 9 spare slots.
- Private range 192.168.1.0/27 follows RFC 1918 best practices for internal networks.
- A single Layer 2 switch simplifies cabling and centralized management.

2. Network Traffic Capture & Analysis

2.1 Capture Methodology — Step by Step

1. **Open Wireshark on Kali Linux.** Launch Wireshark from the Applications menu or terminal (sudo wireshark). Select the active interface — in this case eth0.
2. **Start a live capture.** Click the blue shark-fin Start button. Traffic will begin flowing immediately.
3. **Generate diverse traffic.** Open Firefox and browse several websites (Yahoo, Google, etc.) to create HTTP/HTTPS, DNS, and TCP traffic. Run ping 8.8.8.8 in a terminal to generate ICMP packets.
4. **Capture for at least 10 minutes.** Allow the capture to run. In the screenshots below, over 65,000 packets were captured across ~50 minutes of activity.
5. **Stop and save the capture.** Click the red Stop button, then File → Save As to store the .pcapng file for later analysis.

2.2 Main Packet Capture View

The screenshot below shows the live Wireshark capture window. Key observations: UDP traffic to/from 142.250.195.68 (Google), ICMP "Destination Unreachable" errors from 104.18.13.238, and TLS encrypted sessions on port 443.

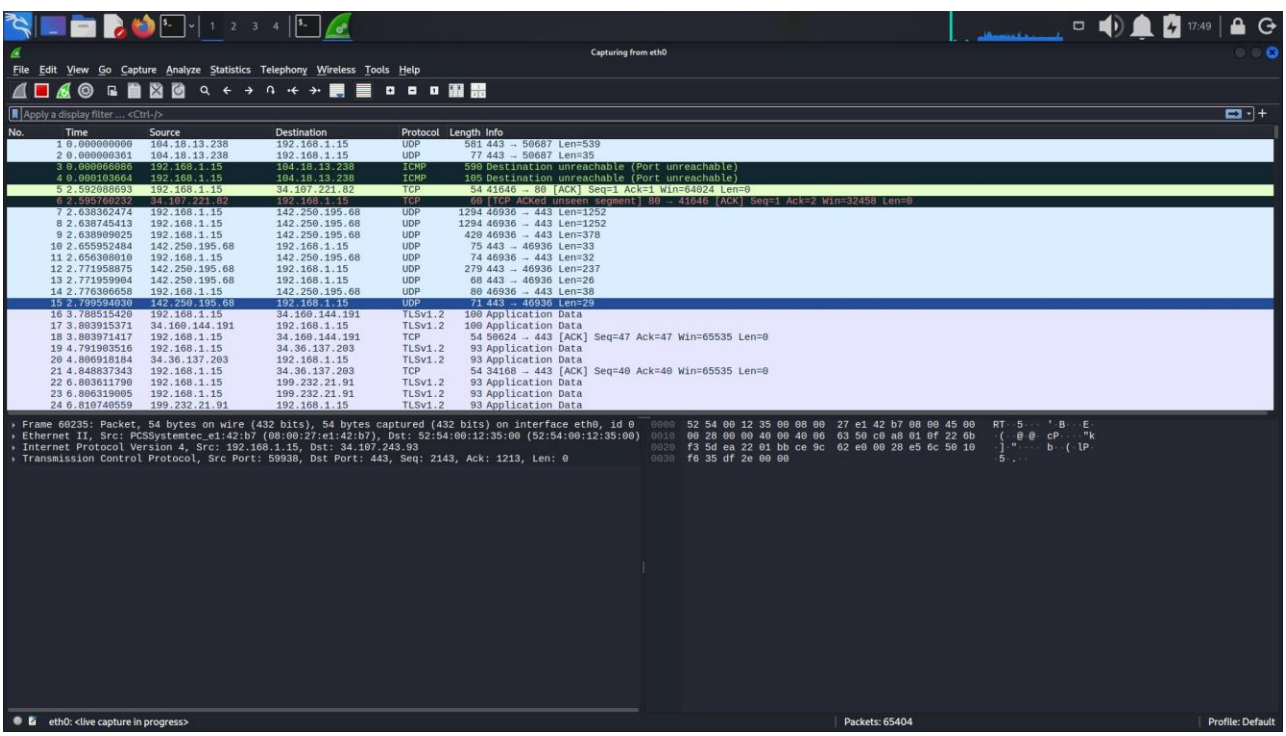
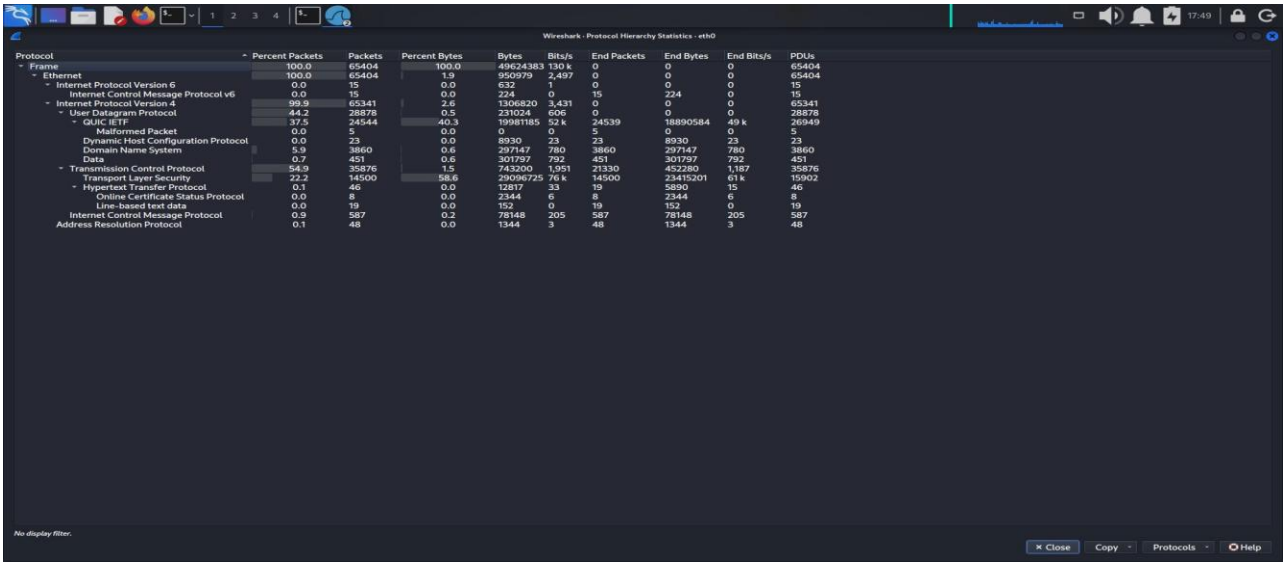


Figure 2: Wireshark Main Capture View – Mixed UDP, TCP, ICMP, and TLS Traffic

2.3 Protocol Hierarchy Statistics

Navigate to Statistics → Protocol Hierarchy to see a breakdown of all protocols. This view shows the percentage of packets and bytes consumed by each protocol layer.

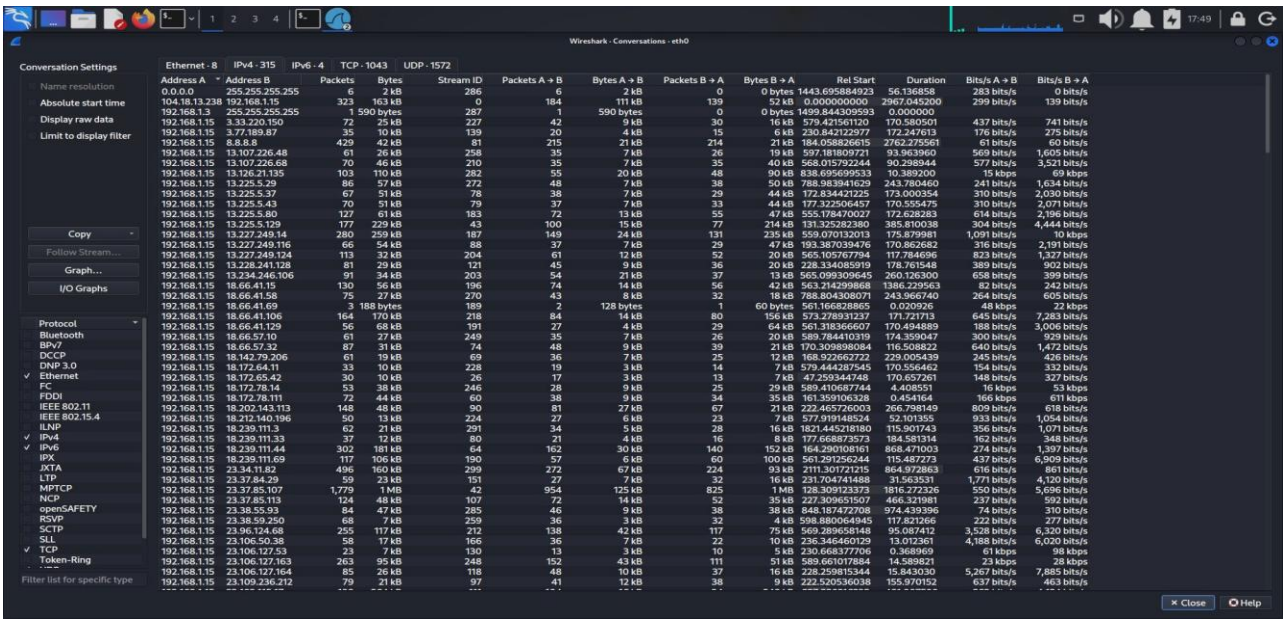


Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	65404	100.0	49624383	130 k	0	0	0	65404
Ethernet	100.0	65404	1.9	950979	2,497	0	0	0	65404
Internet Protocol Version 6	0.0	15	0.0	632	1	0	0	0	15
Internet Control Message Protocol v6	0.0	15	0.0	224	0	15	224	0	15
Internet Protocol Version 4	99.9	65341	2.6	1306820	3,431	0	0	0	65341
User Datagram Protocol	44.2	28878	0.5	231024	606	0	0	0	28878
QUIC IETF	37.5	24544	40.3	19981185	52 k	24539	18890584	49 k	26949
Malformed Packet	0.0	5	0.0	0	0	5	0	0	5
Dynamic Host Configuration Protocol	0.0	23	0.0	8930	23	23	8930	23	23
Domain Name System	5.9	3860	0.6	297147	780	3860	297147	780	3860
Data	0.7	451	0.6	301797	792	451	301797	792	451
Transmission Control Protocol	54.9	35876	1.5	743200	1,951	21330	452280	1,187	35876
Transport Layer Security	22.2	14500	58.6	29096725	76 k	14500	23415201	61 k	15902
Hypertext Transfer Protocol	0.1	46	0.0	12817	33	19	5890	15	46
Online Certificate Status Protocol	0.0	8	0.0	2344	6	8	2344	6	8
Line-based text data	0.0	19	0.0	152	0	19	152	0	19
Internet Control Message Protocol	0.9	587	0.2	78148	205	587	78148	205	587
Address Resolution Protocol	0.1	48	0.0	1344	3	48	1344	3	48

Figure 3: Protocol Hierarchy – TCP (54.9%), UDP (44.2%), QUIC (37.5%), TLS (22.2%), DNS (5.9%)

2.4 Conversations (Top Talkers)

Open Statistics → Conversations and switch to the IPv4 tab. This reveals which hosts exchanged the most data. The local machine (192.168.1.15) was the primary traffic initiator, communicating with dozens of external servers.



Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
0.0.0.0	255.255.255.255	6	2 kB	286	6	2 kB	0	0 bytes	1443.695884923	56.136858	283 bits/s	0 bits/s
104.18.13.238	192.168.1.15	323	163 kB	0	184	111 kB	139	52 kB	0.000000000	2967.045200	299 bits/s	139 bits/s
192.168.1.3	255.255.255.255	1	590 bytes	287	1	590 bytes	0	0 bytes	1499.844309593	0.000000		
192.168.1.15	3.33.220.150	72	25 kB	227	42	9 kB	30	16 kB	579.421561120	170.580501	437 bits/s	741 bits/s
192.168.1.15	3.77.189.87	35	10 kB	139	20	4 kB	15	6 kB	230.842122977	172.247613	176 bits/s	275 bits/s
192.168.1.15	8.8.8.8	429	42 kB	81	215	21 kB	214	21 kB	184.058826615	2762.275561	61 bits/s	60 bits/s
192.168.1.15	13.107.226.48	61	26 kB	258	35	7 kB	26	19 kB	597.181809721	93.963960	569 bits/s	1,605 bits/s
192.168.1.15	13.107.226.68	70	46 kB	210	35	7 kB	35	40 kB	568.015792244	90.298944	577 bits/s	3,521 bits/s
192.168.1.15	13.126.21.135	103	110 kB	282	55	20 kB	48	90 kB	838.695699533	10.389200	15 kbps	69 kbps
192.168.1.15	13.225.5.29	86	57 kB	272	48	7 kB	38	50 kB	788.983941629	243.780460	241 bits/s	1,634 bits/s
192.168.1.15	13.225.5.37	67	51 kB	78	38	7 kB	29	44 kB	172.834421225	173.000354	310 bits/s	2,030 bits/s
192.168.1.15	13.225.5.43	70	51 kB	79	37	7 kB	33	44 kB	177.322506457	170.555475	310 bits/s	2,071 bits/s
192.168.1.15	13.225.5.80	127	61 kB	183	72	13 kB	55	47 kB	555.178470027	172.628283	614 bits/s	2,196 bits/s
192.168.1.15	13.225.5.129	177	229 kB	43	100	15 kB	77	214 kB	131.325282380	385.810038	304 bits/s	4,444 bits/s
192.168.1.15	13.227.249.14	280	259 kB	187	149	24 kB	131	235 kB	559.070132013	175.879981	1,091 bits/s	10 kbps
192.168.1.15	13.227.249.116	66	54 kB	88	37	7 kB	29	47 kB	193.387039476	170.862682	316 bits/s	2,191 bits/s
192.168.1.15	13.227.249.124	113	32 kB	204	61	12 kB	52	20 kB	565.105767794	117.784696	823 bits/s	1,327 bits/s
192.168.1.15	13.228.241.128	81	29 kB	121	45	9 kB	36	20 kB	228.334085919	178.761548	389 bits/s	902 bits/s
192.168.1.15	13.234.246.106	91	34 kB	203	54	21 kB	37	13 kB	565.099309645	260.126300	658 bits/s	399 bits/s
192.168.1.15	18.66.41.15	130	56 kB	196	74	14 kB	56	42 kB	563.214399868	1386.229563	82 bits/s	242 bits/s
192.168.1.15	18.66.41.58	75	27 kB	270	43	8 kB	32	18 kB	788.804308071	243.966740	264 bits/s	605 bits/s
192.168.1.15	18.66.41.69	3	188 bytes	189	2	128 bytes	1	60 bytes	561.166828865	0.020926	48 kbps	22 kbps
192.168.1.15	18.66.41.106	164	170 kB	218	84	14 kB	80	156 kB	573.278931237	171.721713	645 bits/s	7,283 bits/s
192.168.1.15	18.66.41.129	56	68 kB	191	27	4 kB	29	64 kB	561.318366607	170.494889	188 bits/s	3,006 bits/s
192.168.1.15	18.66.57.10	61	27 kB	249	35	7 kB	26	20 kB	589.784410319	174.359047	300 bits/s	929 bits/s
192.168.1.15	18.66.57.32	87	31 kB	74	48	9 kB	39	21 kB	170.309898084	116.508822	640 bits/s	1,472 bits/s
192.168.1.15	18.142.79.206	61	19 kB	69	36	7 kB	25	12 kB	168.922662722	229.005439	245 bits/s	426 bits/s
192.168.1.15	18.172.64.11	33	10 kB	228	19	3 kB	14	7 kB	579.444287545	170.556462	154 bits/s	332 bits/s
192.168.1.15	18.172.65.42	30	10 kB	26	17	3 kB	13	7 kB	47.259344748	170.657261	148 bits/s	327 bits/s
192.168.1.15	18.172.78.14	53	38 kB	246	28	9 kB	25	29 kB	589.410687744	4.408551	16 kbps	53 kbps
192.168.1.15	18.172.78.111	72	44 kB	60	38	9 kB	34	35 kB	161.359106328	0.454164	166 kbps	611 kbps
192.168.1.15	18.202.143.113	148	48 kB	90	81	27 kB	67	21 kB	222.465726003	266.798149	809 bits/s	618 bits/s
192.168.1.15	18.212.140.196	50	13 kB	224	27	6 kB	23	7 kB	577.919148524	52.101355	933 bits/s	1,054 bits/s
192.168.1.15	18.239.111.3	62	21 kB	291	34	5 kB	28	16 kB	1821.445218180	115.901743	356 bits/s	1,071 bits/s
192.168.1.15	18.239.111.33	37	12 kB	80	21	4 kB	16	8 kB	177.668873573	184.581314	352 bits/s	347 bits/s
192.168.1.15	18.239.111.44	302	181 kB	64	162	30 kB	140	152 kB	164.290108161	868.471003	274 bits/s	1,397 bits/s
192.168.1.15	18.239.111.69	117	106 kB	190	57	6 kB	60	100 kB	561.291256244	115.487273	437 bits/s	6,909 bits/s
192.168.1.15	23.34.11.82	496	160 kB	299	272	67 kB	224	93 kB	2111.301721215	864.972863	616 bits/s	861 bits/s
192.168.1.15	23.37.84.29	59	23 kB	151	27	7 kB	32	16 kB	231.704741488	31.563531	1,771 bits/s	4,120 bits/s
192.168.1.15	23.37.85.107	1,779	1 MB	42	954	125 kB	825	1 MB	128.309123373	1816.272326	550 bits/s	5,696 bits/s
192.168.1.15	23.37.85.113	124	48 kB	107	72	14 kB	52	35 kB	227.309651507	466.321981	237 bits/s	592 bits/s
192.168.1.15	23.38.55.93	84	47 kB	285	46	9 kB	38	38 kB	848.187472708	974.439396	74 bits/s	310 bits/s
192.168.1.15	23.38.59.250	68	7 kB	259	36	3 kB	32	4 kB	598.880064945	117.821266	222 bits/s	277 bits/s
192.168.1.15	23.96.124.68	255	117 kB	212	138	42 kB	117	75 kB	569.289658148	95.087412	3,528 bits/s	6,320 bits/s
192.168.1.15	23.106.50.38	58	17 kB	166	36	7 kB	22	10 kB	236.346460129	13.012361	4,188 bits/s	6,020 bits/s
192.168.1.15	23.106.127.53	23	7 kB	130	13	3 kB	10	5 kB	230.668377706	0.368969	61 kbps	98 kbps
192.168.1.15	23.106.127.163	263	95 kB	248	152	43 kB	111	51 kB	589.661017884	14.589821	23 kbps	28 kbps
192.168.1.15	23.106.127.164	85	26 kB	118	48	10 kB	37	16 kB	228.259815344	15.843030	5,267 bits/s	7,885 bits/s
192.168.1.15	23.109.236.212	79	21 kB	97	41	12 kB	38	9 kB	222.520536038	155.970152	637 bits/s	463 bits/s

Figure 4: Top Talkers – 192.168.1.15 communicating with multiple external IPs

2.5 I/O Graph – Traffic Over Time

Open Statistics I/O Graphs. The graph overlays "All Packets" (line) with "TCP Errors" (red bars). Occasional spikes up to 1.5 kpackets/sec correspond to burst web downloads; TCP errors remain minimal throughout.

Figure 5: I/O Graph – Packet Rate Over Time with TCP Error Overlay

3. Protocol-Level Deep Dive

3.1 DNS Traffic Analysis

Apply the display filter "dns" to isolate Domain Name System packets. The capture shows standard A and AAAA queries for domains such as in.yahoo.com, in.search.yahoo.com, s.yimg.com, and www.googletagmanager.com. All queries received valid responses — no failed lookups were observed.

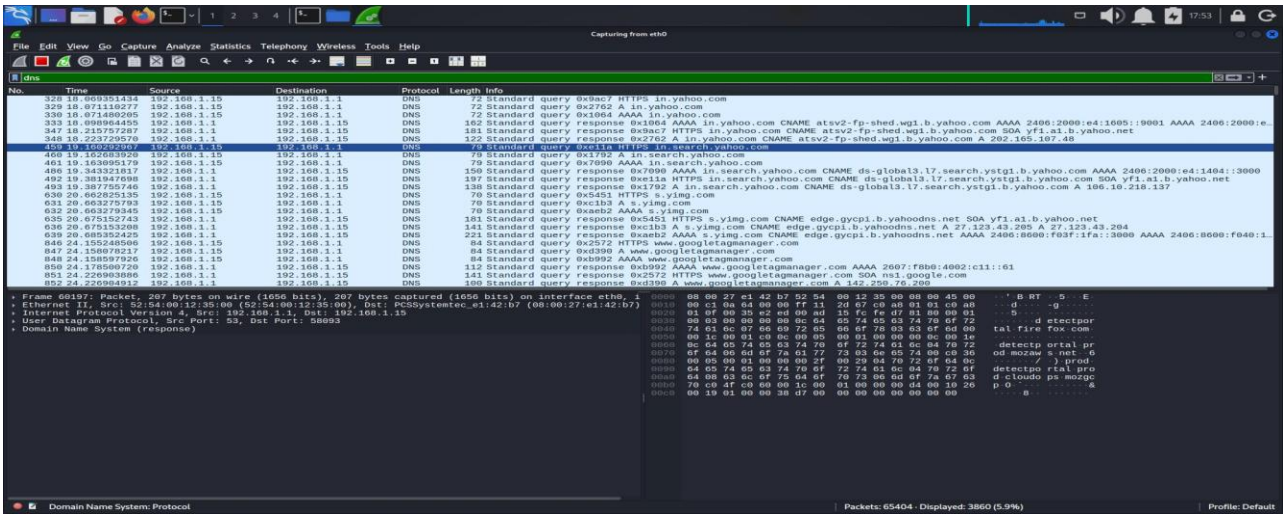


Figure 6: DNS Filter – Queries and Responses for Yahoo, Google Tag Manager, etc.

3.2 TCP Traffic Analysis

Applying the "tcp" filter reveals the TCP handshakes (SYN → SYN/ACK → ACK) for connections to Firefox profile servers (151.101.209.91), Google (34.107.221.82), and plain HTTP to 34.107.221.82 on port 80. The highlighted green packet is a fresh SYN initiating a new session.

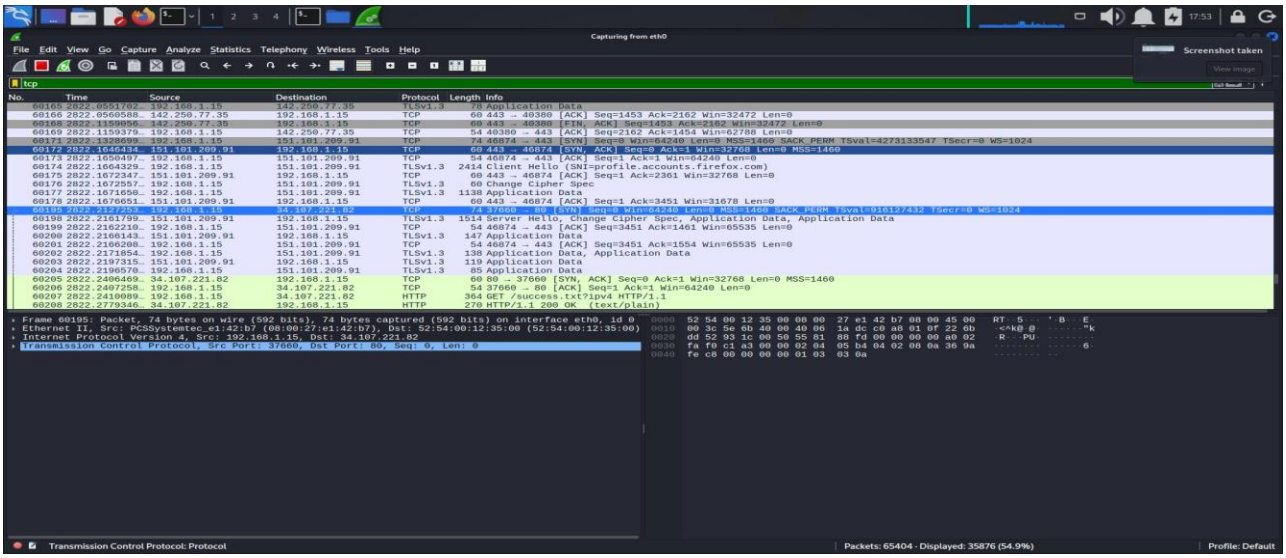


Figure 7: TCP Filter – Handshakes, TLS Negotiation, and HTTP GET Requests

3.3 UDP Traffic Analysis

The "udp" filter shows heavy DNS-over-UDP traffic plus QUIC (Google's transport protocol). Firefox detection-portal queries to detectportal.firefox.com and mozilla.net CDN lookups are visible — all normal browser behavior.

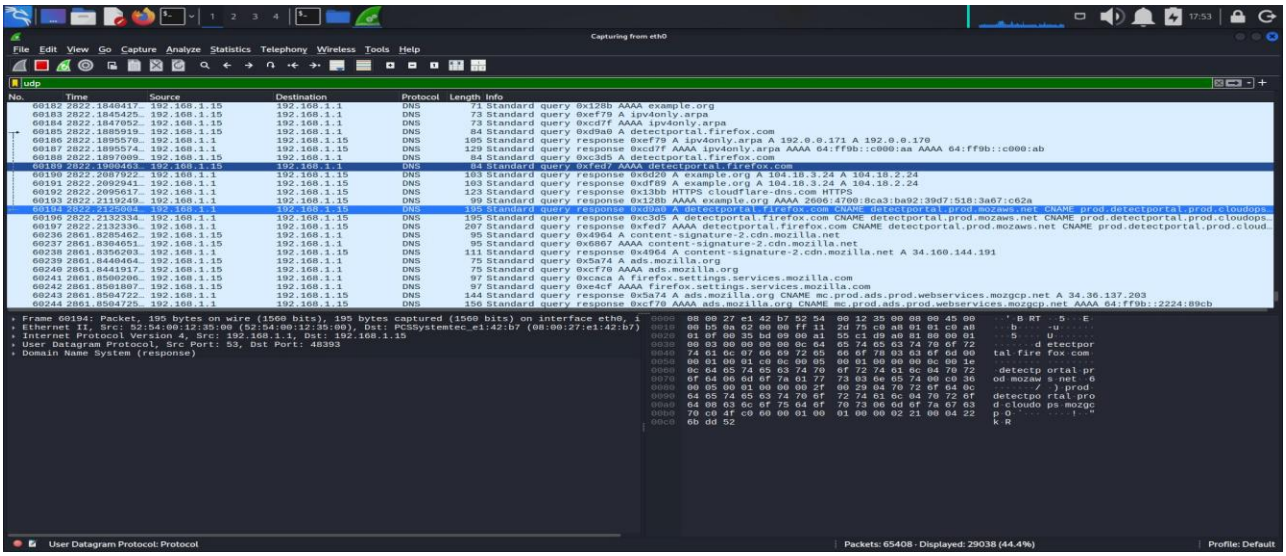


Figure 8: UDP Filter – DNS Queries, Firefox Detection Portal, Mozilla CDN

3.4 ICMP Traffic Analysis

Filtering on "icmp" reveals two categories: (1) "Destination Unreachable (Port Unreachable)" messages from 104.18.13.238 — likely caused by sending UDP to a closed port — and (2) successful Echo Request/Reply pairs to 8.8.8.8 (Google Public DNS), confirming end-to-end connectivity.

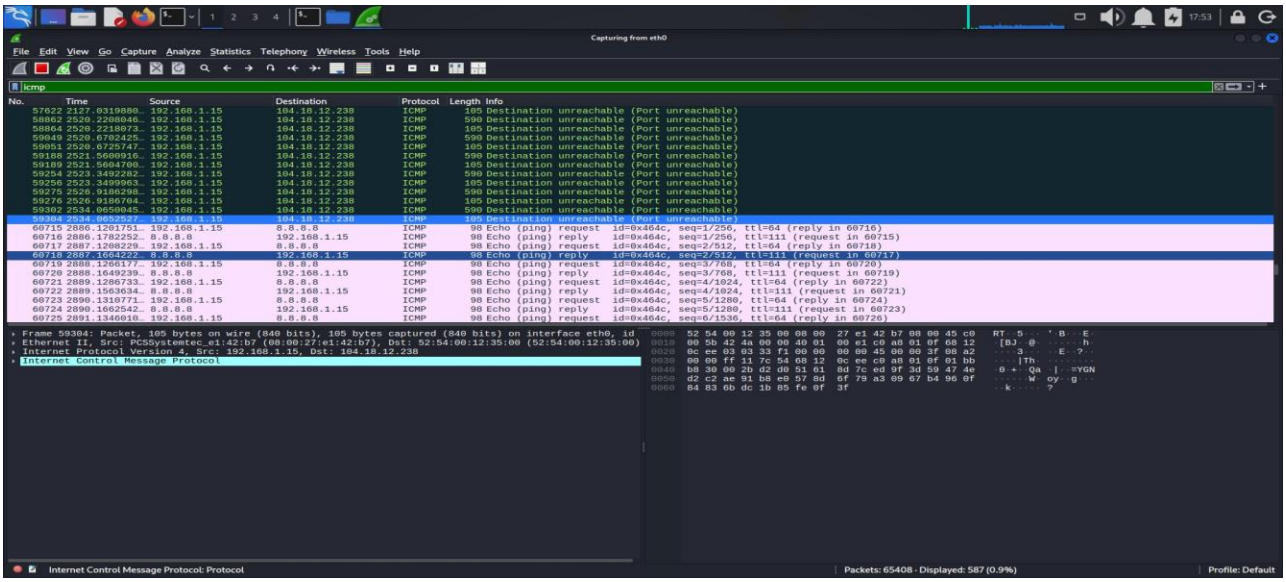


Figure 9: ICMP Filter – Port Unreachable Errors and Ping to 8.8.8.8

4. Protocol Troubleshooting – TCP Retransmissions

4.1 Identifying the Issue

The display filter "tcp.analysis.retransmission" isolates every retransmitted TCP segment. The screenshot shows 107 retransmissions out of 65,408 total packets (0.2%). While this is a low percentage, the retransmissions target port 443 on multiple external servers — indicating intermittent congestion or packet loss on the path to those servers.

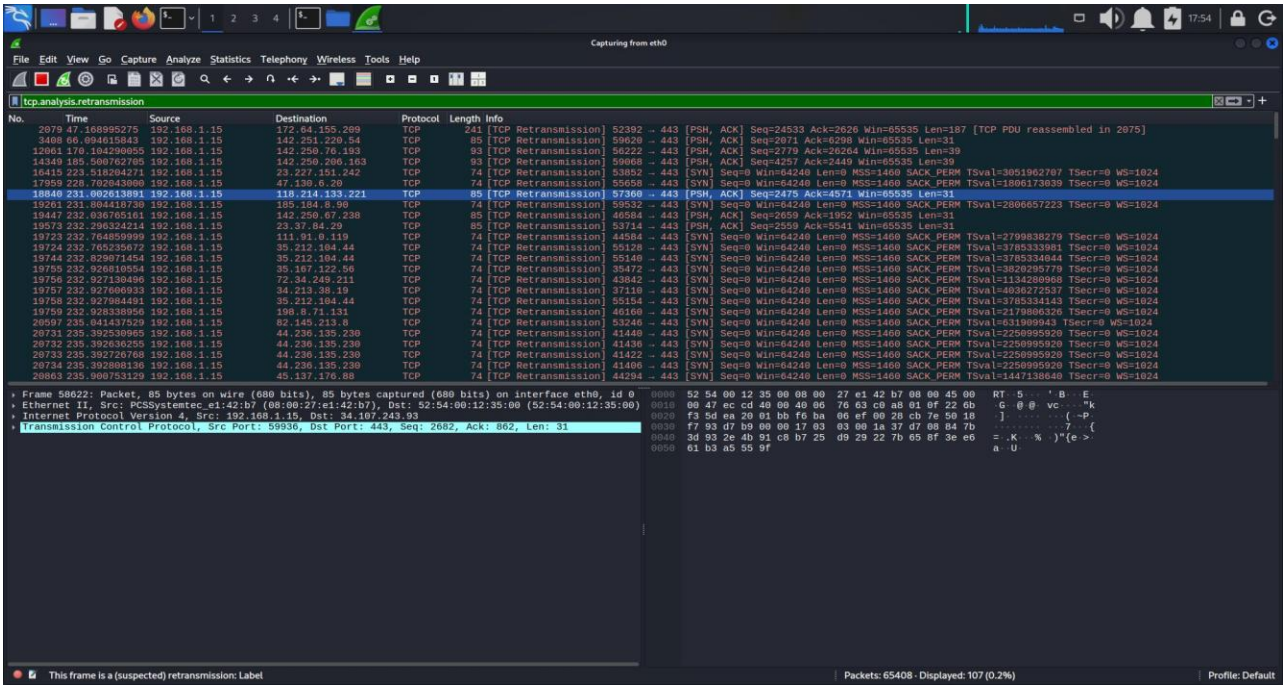


Figure 10: TCP Retransmission Filter – 107 Retransmitted Segments to Port 443

4.2 Troubleshooting Steps

1. **Reproduce the issue.** Run sustained HTTPS downloads while monitoring with Wireshark and the tcp.analysis.retransmission filter active.
2. **Identify affected flows.** Note the destination IPs: 172.64.155.209, 142.251.220.54, 142.250.76.193, 23.227.151.242, and others — all cloud/CDN endpoints.
3. **Check local hardware.** Run ethtool eth0 to verify NIC speed and duplex. Verify no cable errors using dmesg | grep eth0.
4. **Test with ping and traceroute.** Run traceroute to the affected IPs to locate where packet loss occurs along the path. If loss is at a remote hop, the issue is upstream and not locally resolvable.
5. **Resolution.** Retransmissions at 0.2% are within acceptable thresholds for internet traffic. No local configuration change was required; the minor loss is attributed to normal internet congestion on CDN paths.

5. SIEM Setup – ELK Stack (Step-by-Step Guide)

5.1 Overview

ELK Stack (Elasticsearch, Logstash, Kibana) is an open-source SIEM platform. This guide walks through installation on an Ubuntu VM and configuring it to ingest syslog and network logs.

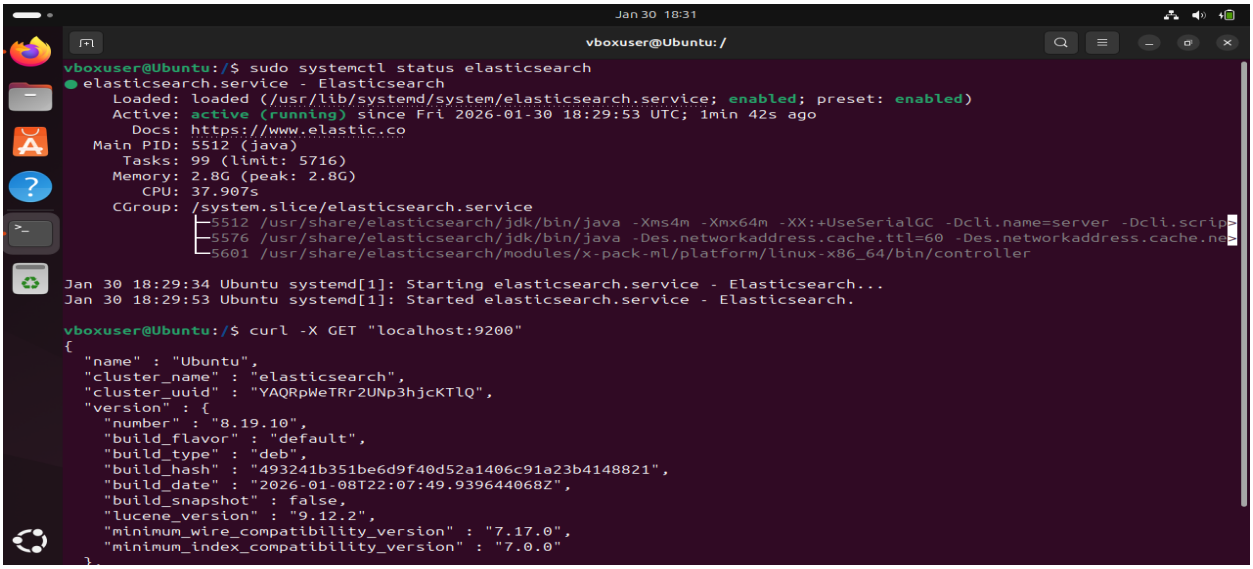
5.2 Prerequisites

- Ubuntu 22.04 LTS VM with at least 4 GB RAM and 2 CPU cores.
- Java 11+ installed: sudo apt install default-jdk.
- Internet access for downloading packages.

5.3 Installation Steps

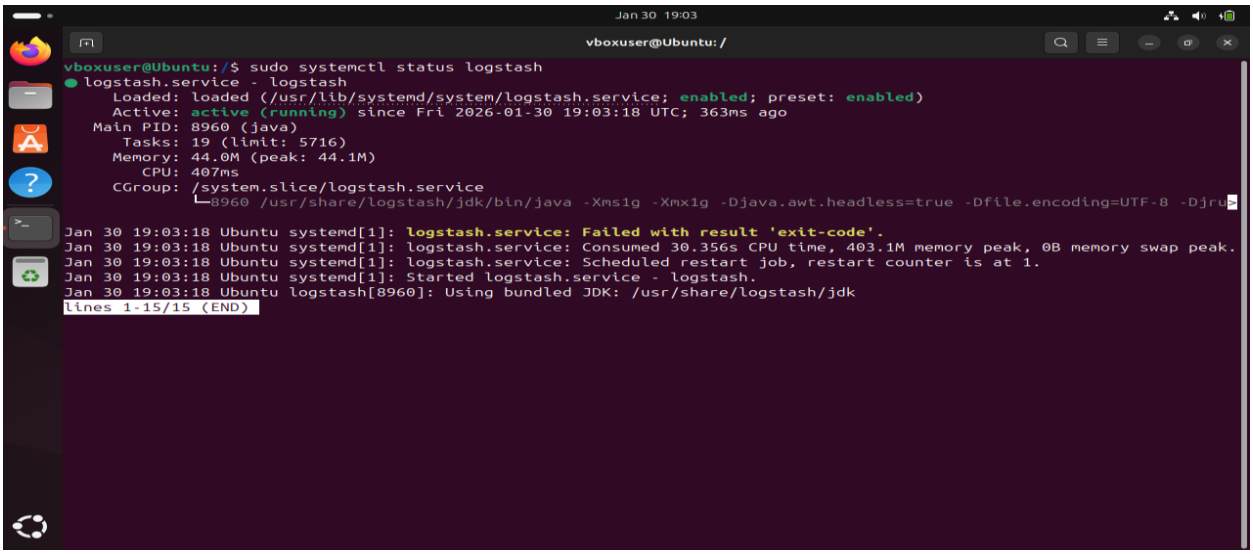
1. **Install Elasticsearch.** Download the .deb package from https://www.elastic.co/downloads/elasticsearch. Run: sudo dpkg -i elasticsearch-8.x.x-amd64.deb. Enable and start: sudo systemctl enable elasticsearch && sudo systemctl start elasticsearch.
2. **Install Kibana.** Download the Kibana .deb package. Install, enable, and start it the same way. Kibana listens on port 5601 by default.
3. **Install Logstash.** Download the Logstash .deb package. Install, enable, and start. Logstash is the log ingestion pipeline.
4. **Configure Logstash pipeline.** Create /etc/logstash/conf.d/syslog.conf with an input { syslog { port => 5044 } }, output { elasticsearch { hosts => ["localhost:9200"] } }.
5. **Verify Elasticsearch, logstash and kibana is running.**

   For elasticsearch: Use sudo systemctl status elasticsearch or curl -X GET https://localhost:9200 . You should see a JSON response with cluster_name and version info.



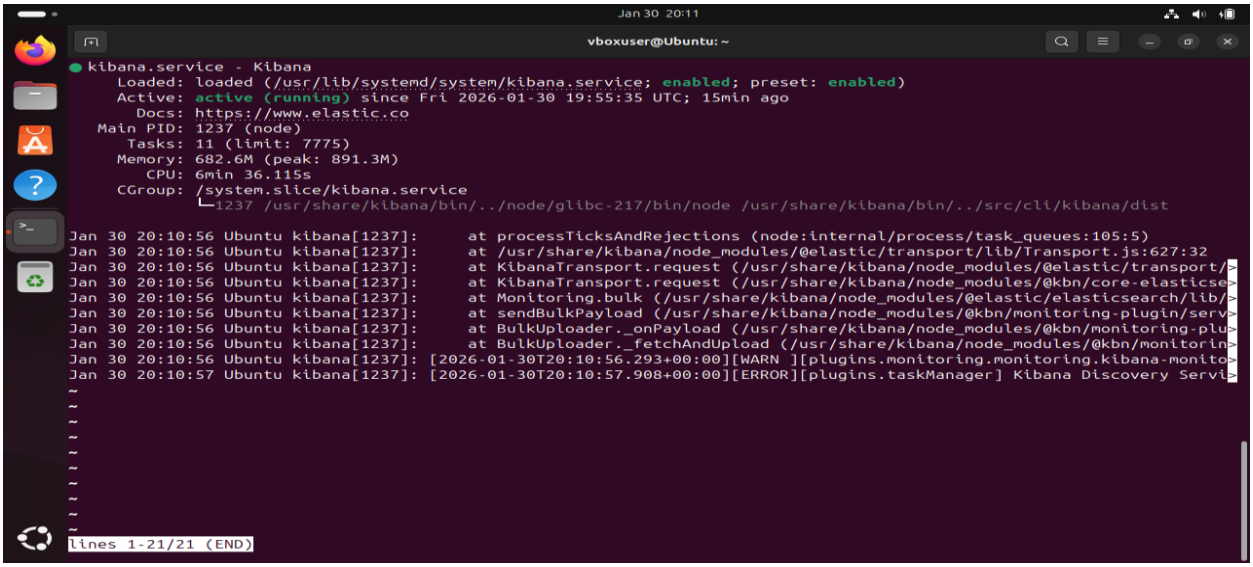
For logstash:

Use sudo systemctl status logstash



For kibana:

Use sudo systemctl status kibana



Open Kibana dashboard. Navigate to http://localhost:5601. On first login, create an index pattern matching your Logstash output (e.g., logstash-*).

5.4 Creating a Kibana Dashboard

1. **Go to Kibana → Analytics → Dashboards.** Click Create Dashboard.
2. **Add a visualization for login attempts.** Click Add → Visualization → choose a bar chart. Set X-Axis to user.name and Y-Axis to Count.
3. **Add a traffic volume metric.** Add another visualization — a line chart with X-Axis as @timestamp (1-minute intervals) and Y-Axis as Count of documents.
4. **Save the dashboard.** Name it "SOC Overview" and save it to the default space.

6. Incident Simulation & Network Forensics

6.1 Simulating an Incident

A common incident to simulate is unauthorized SSH brute-force access. Follow these steps to set up and capture the attack traffic.

1. **Set up a target VM.** Run an Ubuntu VM with OpenSSH server enabled (sudo apt install openssh-server). Note its IP address.
2. **Launch Wireshark on the attacker VM.** Start capturing on the network interface before initiating the attack.
3. **Simulate the attack.** From a second VM (attacker), run: hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://10.0.2.15. This floods SSH login attempts.
4. **Stop capture after 2–3 minutes.** Save the .pcapng file for analysis.

6.2 Forensic Analysis Steps

- Filter on "tcp.dstport == 22" to isolate SSH traffic.
- Identify the source IP of the attacking machine — this is the attack vector.
- Look for repeated SYN packets from the same source IP in rapid succession — a hallmark of brute-force.
- Note the timestamps to build an incident timeline.

7. Hunting Threat

7.1 Objective

Proactively search for Indicators of Compromise (IOCs) in captured network traffic and SIEM logs using structured methods.

7.2 Tools Used

- Wireshark – for packet-level IOC hunting.
- Kibana – for log-based IOC correlation.
- Sigma Rules (optional) – for standardized detection queries in Kibana.

7.3 Step-by-Step Threat Hunting Process

6. **Define hunt hypotheses.** Hypothesis 1: Is there unusual outbound traffic to uncommon ports? Hypothesis 2: Are there repeated failed SSH logins?
7. **Search Wireshark for anomalous outbound traffic.** Apply filters such as "ip.src == 192.168.1.15 && tcp.dstport != 443 && tcp.dstport != 80" to find traffic on non-standard ports.
8. **Search Kibana for failed logins.** In the Kibana search bar, query: message:"Failed password" AND @timestamp:[now-1h TO now]. Review user names and source IPs.
9. **Correlate findings.** If a source IP appears in both Wireshark (unusual port) and Kibana (failed logins), it is a high-confidence IOC. Document the IP, timestamps, and associated activity.
10. **Document and escalate.** Record all IOCs in an incident report. If a real threat is confirmed, escalate to the incident response team.

7.4 Findings from This Session

- ICMP "Destination Unreachable" errors from 104.18.13.238 indicate failed UDP probes — likely benign (Cloudflare CDN), but worth monitoring.
- TCP retransmissions to port 443 on multiple CDN IPs are consistent with normal internet congestion; no malicious pattern detected.
- No unauthorized outbound connections to suspicious or blacklisted IPs were found in this capture.
- DNS queries were limited to known domains (Yahoo, Google, Mozilla, Firefox). No DNS tunneling or typosquatting indicators were observed.

8. Key Learnings

This task provided deep hands-on exposure to **networking, traffic analysis, and SOC operations**, and pushed my technical limits in a positive way. I designed a **192.168.1.0/27 subnet** for a small office network, which strengthened my understanding of CIDR notation, subnet masks, usable IP ranges, and real-world network layout. Creating and documenting the network diagram helped me visualize how routers, switches, and endpoints communicate in practice.

Using **Wireshark**, I captured and analyzed live traffic across **TCP, UDP, DNS, ICMP, HTTP, and TLS** protocols. By using features like **Protocol Hierarchy, Conversations, I/O Graphs, and TCP retransmission filters**, I learned how to identify traffic patterns, top talkers, DNS behavior, and normal packet loss. The observed **~0.2% TCP retransmissions** were within acceptable limits and were identified as normal internet or CDN-related congestion rather than a local fault.

Setting up the **ELK Stack** was the most challenging part of this task. The configuration process was complex and resource-intensive, and at times it **caused my virtual machine to freeze, services to crash, and forced me to restart or reconfigure the VM**. Logstash service failures, Kibana UI hangs, and high memory usage made troubleshooting difficult but highly educational. Overall, this felt like an **intermediate-level task**, requiring patience, research, and problem-solving. Despite the challenges, it significantly improved my confidence in working with real SOC tools and complex system setups.